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Artificial intelligence driven sustainable management of plant resources: From cultivation to commercialization

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Abstract-Artificial intelligence (AI) presents a transformative paradigm for sustainable management of plant resources in agriculture. In this paper, we review how AI technologies are transforming the entire value chain of plant resources (e.g., precision cultivation, resource optimization and post-harvest management, and market commercialization). Machine learning, coupled with computer vision and predictive analytics, can power AI systems to deliver increased crop yields, minimal environmental impact and better financial returns for all of the stakeholders involved in agriculture.

Keywords: Artificial intelligence, Sustainable management, Plant Resources, environmental impact

INTRODUCTION

Global agriculture brings a series of challenges like climate change, rapidly increasing population, resource depletion and problems of food security.[1] Conventional farming practices are unable to keep pace with these requirements in a sustainable way. AI provides modern revolutionary solutions by analyzing huge databases, recognizing patterns and supporting data-based decision-making throughout various plant production processes.[2] This review collates the existing knowledge on AI assisted sustainable management of plant resources from cultivation to commoditization and provides an overview for technological interventions.

AI in Precision Cultivation

Precision agriculture and its sensors are reaping the benefits of Artificial Intelligence for crop monitoring, irrigation control and soil management. Intelligent crop monitoring based on computer vision and deep learning supports with more than 95% accuracy the early detection of diseases, leading to timely reactions and reduction of pesticide volumes.[3] AI-driven drones and satellites also offer real-time crop assessment to detect stress before it is visible, enabling targeted agrochemical use and curbing harmful-environmental exposure. Smart irrigation systems use machine learning and AI to determine precisely when to water based on soil moisture, weather and the needs of the crop. These systems may reduce water use by 20-30% without reducing or even increasing yields.[4] For soil health management, AI models analyze soil composition and nutrients to provide accurate fertilization schedules. They predict nutrient deficit, therefore decreasing fertilizer waste and groundwater contamination, make farming more sustainable and efficient.

AI in Crop Selection and Breeding

Artificial intelligence (AI) speeds up plant breeding programs by using genomic data to forecast phenotypes.

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Genetic markers related to important traits, for example drought tolerance, disease resistance and nutritional value are revealed using machine-learning approaches.[5] At the end, with accelerated use of AI-based breeding programs potentially cutting decades off the development time for new cultivars, rapid adaptation to changing environmental conditions and consumer tastes may be achieved. Predictive models also help farmers choose crops by using data for climate, soil and market station. These decision-support systems maximize land productivity, minimize environmental impact and optimize crop rotation strategies by optimal allocation to resources.[6]

AI in Sustainable Pest and Weed Management

Sustainable agriculture management is transformed through artificial intelligence for pest, weed and post-harvest practices. Intelligent pest control Automated pests detection and elimination Process uses AI image recognition and computer vision to accurately identify infestations, thus providing targeted interventions. Robots can apply pesticides or implement mechanical controls only where needed, which results in a reduction of nearly 90% in pesticide use over conventional methods while saving natural enemies.[7]

Smart weed control utilizes machine learning algorithms to determine the difference between a crop and weed, and then to apply an herbicide or mechanically remove the unwanted plant. These systems dramatically reduce herbicide application, reducing environmental contamination and financial costs - making them particularly useful for organic farming.[8]

Post-Harvest Management and Quality Control

Post-harvest management is supported by AI computer vision to automatically sort and grade the produce according to size, color, ripeness. Deep learning helps in maintaining the uniform quality, decreases human reliance and also reduces food wastage.[9] Machine learning enabled predictive storage management regulates temperature, humidity and atmosphere on the fly, leading to reduction in spoilage upto 30% of food lost globally post-harvest. These tools do away with post-harvest losses (estimated at 30%, worldwide) that mean vast economic and environmental waste.

AI in Supply Chain Optimization and Commercialization

AI also supports supply chain optimization with demand prediction as well as logistics management. Predictive analytics can predict consumer behaviour, in this way the production is synchronized not to produce more than required.[10] Transportation routes and storage allocation can also be optimized to reduce carbon emissions and spoilage using machine learning. The combination of AI and blockchain generates transparency, because it confirms the sustainable cultivation and therefore product authenticity.[11] AI applications In economic and environmental terms, 15-25% of costs of agricultural input can be reduced by AI systems, yield can go up 10-30%, and ecological impacts can be minimal.[2] The value of the world market for AI in agriculture is estimated to achieve $4.7 billion by 2028, demonstrating its transformative potential for sustainable food systems.[1]

Table 1- AI Applications Across Plant Resource Management Stages

Stage	AI Technology	Sustainability Benefit	Efficiency Gain
Cultivation	Computer vision, ML sensors	30% reduction in water use	20-25% yield increase
Pest Management	Image recognition, robotics	90% reduction in pesticides	15% cost reduction
Disease Detection	Deep learning, CNNs	Early intervention, reduced crop loss	95% detection accuracy
Breeding	Genomic analysis, predictive modelling	Climate-adapted varieties	50% faster development
Post-Harvest	Quality assessment, storage optimization	30% reduction in spoilage	20% labor cost reduction
Commercialization	Demand forecasting, logistics optimization	Reduced food waste, lower emissions	15% distribution efficiency

Challenges and Future Directions

Although there are promising use cases for such an application, AI implementation in agriculture is hampered by some major challenges including cost barriers, scarce digital infrastructure in rural areas and limited technical capacity among the farmers. Considering issues such as data privacy and algorithm transparency is critical to ensure fair technology access for all, and protect against smallholder exploitation. To this end, further studies should concentrate on designing cost-effective, farmer-friendly AI-based tools suited to different farming communities. By applying cutting-edge technologies such as edge

computing, 5G networks and advanced robotics, the AI will be developed in terms of sustainable plant production.

CONCLUSION

Artificial intelligence is an innovative solution for sustainable management of plant resources from planting to market release. AI enabled solutions improve resource utilization and minimize the impact on the environment, while maximizing economic viability throughout the entire agricultural value chain. Successful integration of eVoting should consider how to overcome barriers to adoption, provide fair access and sustained focus on sustainability goals. As AI applications develop and new solutions bring technology closer to users, the ability of AI to help feed the world in a way that minimizes harm to natural resources is likely to grow.

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